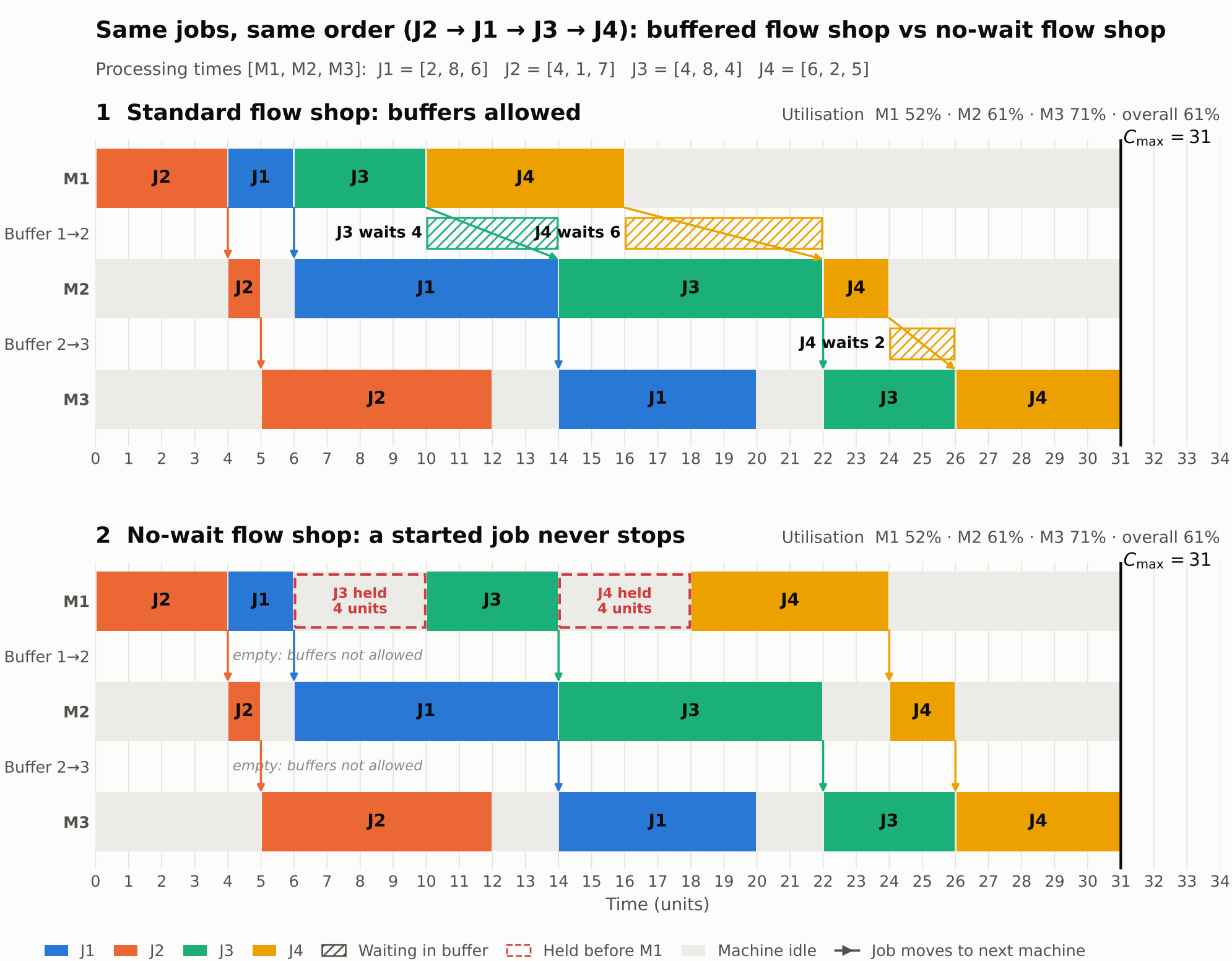
Same jobs, same order (J2 → J1 → J3 → J4): buffered flow shop vs no-wait flow shop
Processing times [M1, M2, M3]: J1 = [2, 8, 6] J2 = [4, 1, 7] J3 = [4, 8, 4] J4 = [6, 2, 5]
1 Standard flow shop: buffers allowed
Utilisation M1 52% · M2 61% · M3 71% · overall 61%
$C_{max} = 31$
M1
Buffer 1→2
M2
Buffer 2→3
M3
J2
J1
J3
J4
J3 waits 4
J4 waits 6
J4 waits 2
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
2 No-wait flow shop: a started job never stops
Utilisation M1 52% · M2 61% · M3 71% · overall 61%
$C_{max} = 31$
J3 held 4 units
J4 held 4 units
empty: buffers not allowed
empty: buffers not allowed
Time (units)
J1
J2
J3
J4
Waiting in buffer
Held before M1
Machine idle
Job moves to next machine